Reading Environment and History through Tree Rings

Lesson 3 Principles of dendrochronology · 90 minutes

Crossdating tests dates for individual rings by matching shared patterns among series. Distinguish this from simply counting rings.

Interpreting shared variation

Growth reflects age and size, climate, competition and disturbance. Limiting factors depend on species, site and period.

Replication can involve different radii, trees or sites. Two radii from each of two trees yield four samples, not four independent trees.

Uniformitarianism is a starting point; growth responses need not remain constant. Check dating and environmental interpretation separately.

Matching procedure

1. Check boundaries and ring numbers.

2. Compare patterns with the reference using equal annual spacing.

3. Compare alternative placements and investigate mismatches.

4. Check anatomy, other radii, other trees and statistical evidence.

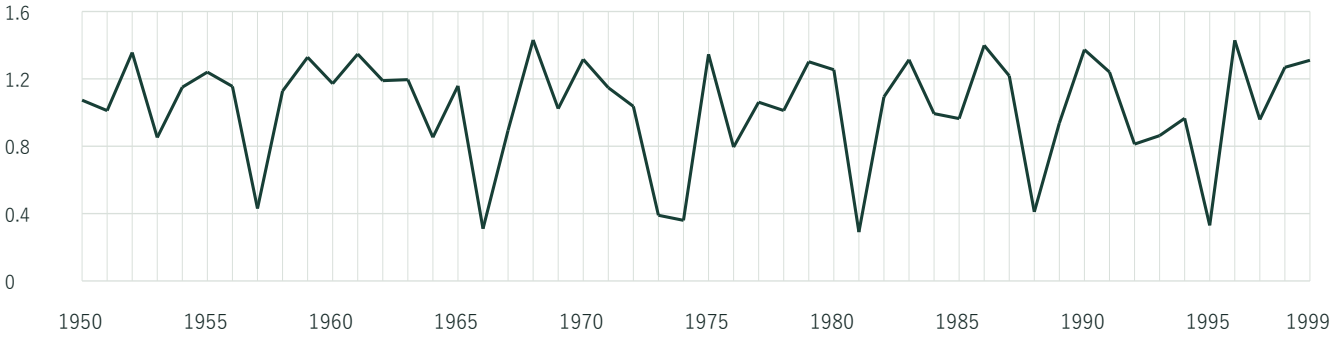
5. Record reasoning and unresolved questions.

All exercise data are synthetic arbitrary units, not empirical ring widths. Short sequences around 30 rings practise a procedure and do not define dating criteria.

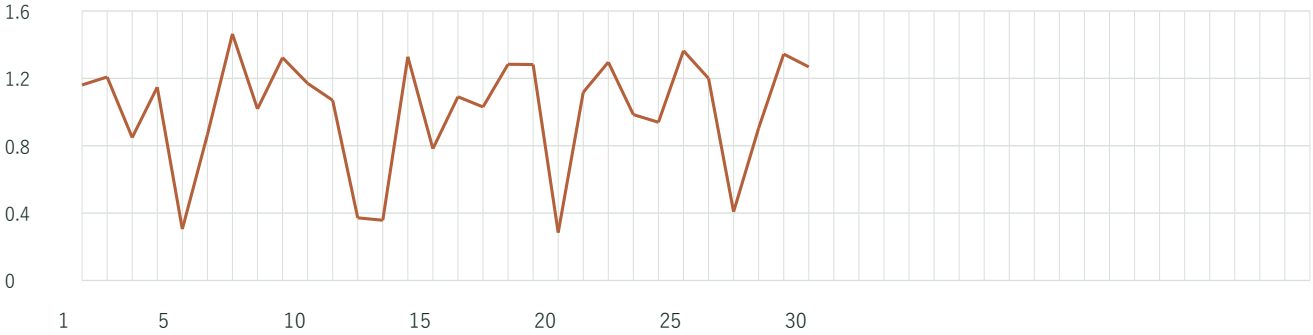
Exercise A A complete 30-ring sequence

Place A within reference M, 1950–1999, and propose first and last years. Both plots use the same annual spacing. Cut-out strips are on page 6.

M: 1950–1999



A: ring number 1–30



Proposed first / last year:

Record at least three matching features. Distinguish ring numbers from candidate calendar years.

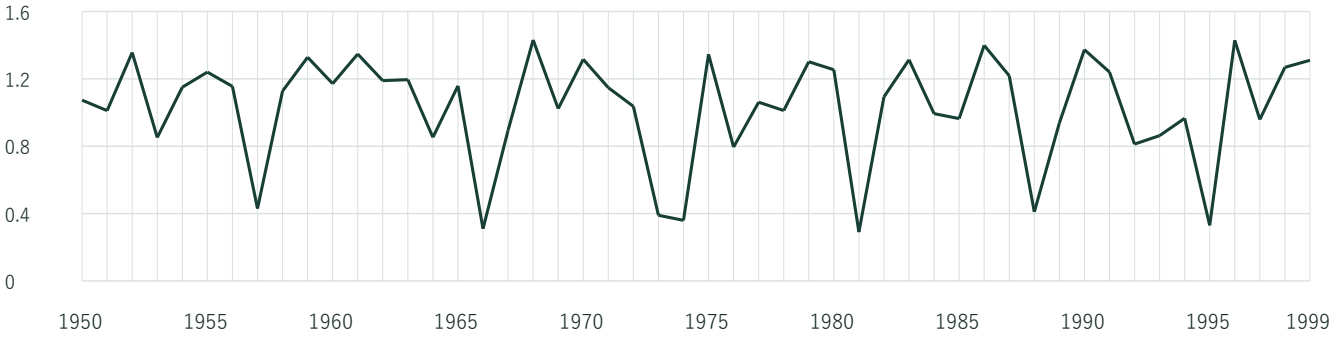
Ring number in A	Candidate year in M	Matching feature

Where does an alternative placement fail?

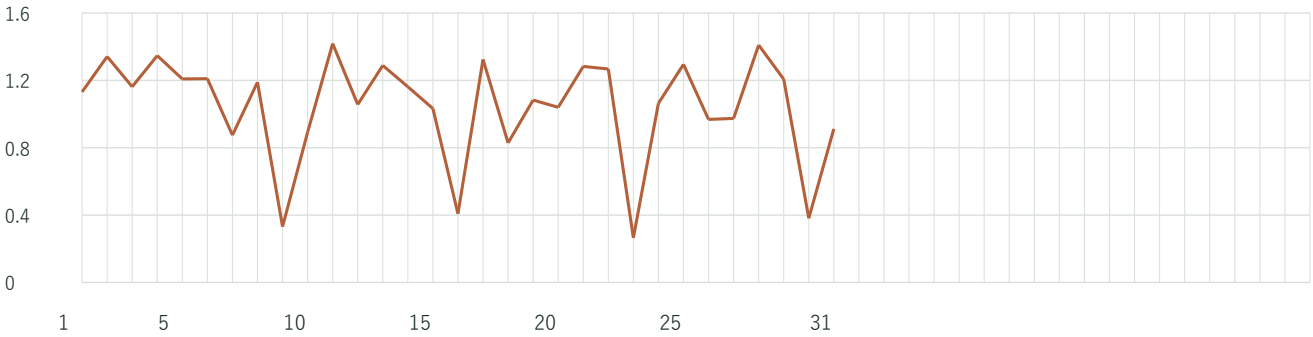
Exercise B A shift within 31 observed rings

Compare early and late segments. Can one constant shift align the whole sequence? Identify where to re-examine the specimen before deciding the cause.

M: 1950–1999



B: ring number 1–31



Candidate start for the early segment / later shift:

Ring positions surrounding the shift and possible causes:

What evidence could distinguish a locally absent increment from a missed boundary?

Verification and links to the final report

Correlation helps narrow candidate placements. Interpretation depends on overlap, preprocessing, autocorrelation and search range. One high coefficient is insufficient.

Agreement with a reference already containing the unknown is not an independent test. Use a reference excluding it or other samples.

An inferred missing increment is not an observed width of zero. Preserve original data and record missing positions separately.

Record your decision

Conclusion and supporting evidence:

Mismatches or remaining uncertainty:

Additional specimens, observations or analyses needed:

Preparing the final report

For a research plan or review, distinguish the question, evidence and method, results or expected results, and limitations or alternatives. The syllabus specifies 2–4 A4 pages, optional figures, and at least three references; deadlines and marking details will be announced.

Terms and linked resources

Japanese / English	Brief explanation
制限因子（limiting factor）	An environmental or physiological constraint on growth.
反復試料（replication）	Checking reproducibility across radii or trees.
クロスデーティング（crossdating）	Verifying ring dates through shared patterns.
標準年輪曲線（reference chronology）	A securely dated series used as a reference.
浮動年輪曲線（floating chronology）	An internally dated sequence not anchored to calendar years.
スケルトンプロット（skeleton plot）	Marks representing relatively narrow or distinctive rings.
重複年数（overlap length）	Number of years shared by two sequences.
局所的欠損輪（locally absent ring）	An annual increment absent from part of the stem.

R21 Instructional crossdating exercises

R22 Making skeleton plots

R23 The dated master chronology

R24 Finding the dates

R25 Absent and false rings

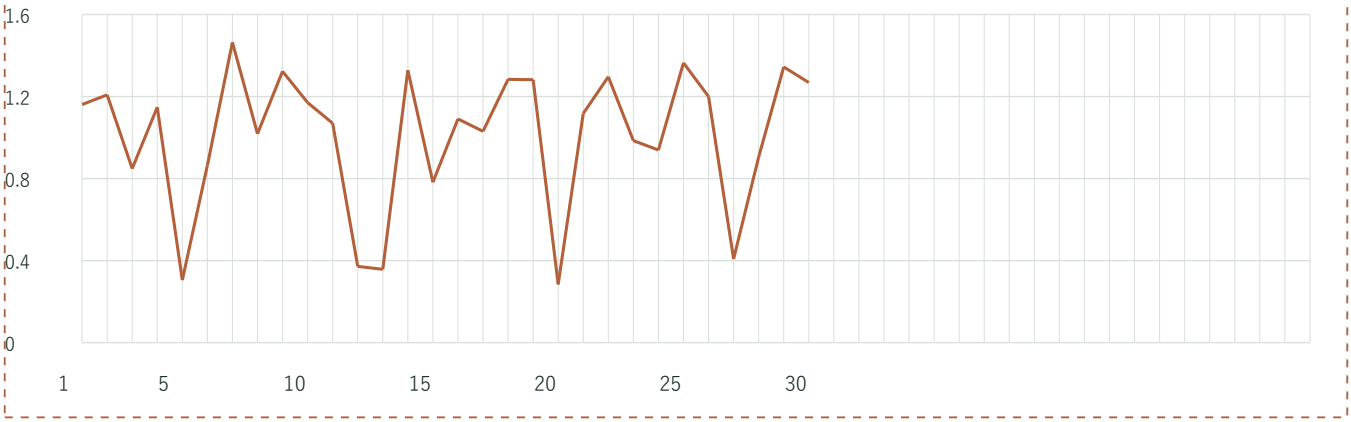
R08 International Tree-Ring Data Bank

Search terms: crossdating / marker year / skeleton plot / missing ring / reference chronology.

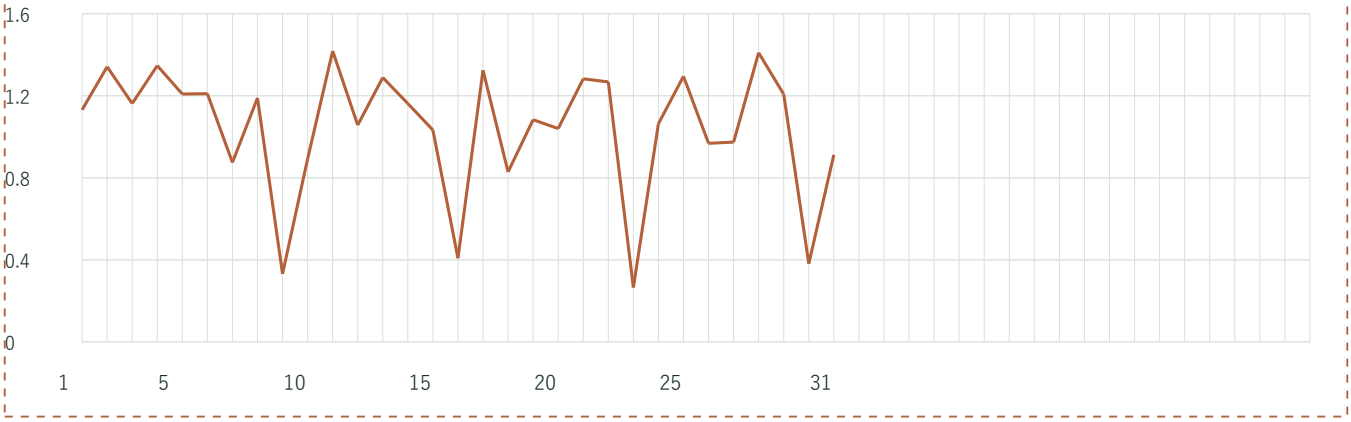
Cut-out curve strips

Cut around A and B. Move each strip below M on pages 2–3, keeping the same horizontal step. Match several features rather than just endpoints.

A: ring number 1–30



B: ring number 1–31



Optional spreadsheet activity

Create an XY scatter plot with straight lines from the supplied CSV columns. A and B use ring numbers; M uses calendar years. In a working copy, add candidate start year + ring number − 1. Keep missing-ring hypotheses in separate columns and preserve source CSV files.

Lesson 3 Make a skeleton plot from measurements

About eight minutes, replacing the synthetic Exercise A on slides 14–16. These are the innermost 15 measured rings of well-frame boards A1 (29ho018) and C5 (29ho017), published by Nabunken. Equal ring indices do not imply equal calendar years.

Ring index	29ho018 (0.01 mm)	29ho017 (0.01 mm)
1	32	67
2	44	85
3	42	69
4	52	74
5	47	83
6	39	124
7	23	98
8	65	147
9	75	128
10	52	114
11	42	97
12	49	90
13	44	75
14	34	30
15	34	63

Mark conspicuously narrow rings in 29ho018. Draw equally spaced indices 1–15 and longer vertical marks for relatively narrower rings. Repeat for 29ho017 and compare the spacing between features.

Do not use measured width directly as mark length. Omit ordinary rings. This 15-ring excerpt teaches the procedure; it is not sufficient evidence for dating.

Measured data: Nabunken 29ho018, CC BY 4.0

Comparison specimen: 29ho017, CC BY 4.0

Lesson 3 Why withhold a date?

Example interpretation

In 29ho018, ring 7 (0.23 mm) is conspicuously narrow. Rings 14 and 15 (0.34 mm each) are narrower than ring 13 (0.44 mm), but the excerpt does not show the outer neighbour of ring 15. In 29ho017, ring 14 (0.30 mm) stands out. Mark lengths reflect relative judgements, so there is no unique numerical answer.

A partial match in a short excerpt does not establish a calendar date. Examine the full series, images and a securely dated reference. The records describe separate boards, but do not establish that they came from separate trees. Separate files are not necessarily independent biological replicates.

Read the Yoshino specimen coverage (slide 25)

Calendar year	Specimens present	Measured radii
1636	1	1
1650	2	3
1700	6	7
1800	24	30
2000	57	69
2018	13	18

How does replication differ between 1636 and 1800? Should two radii from the same tree count as two independent trees?

Counts are recalculated from the first and last years of specimens associated with the published study. Coverage describes replication, not a direct measure of dating correctness or reconstruction skill.

Yonenobu et al. (2026): Yoshino cedar